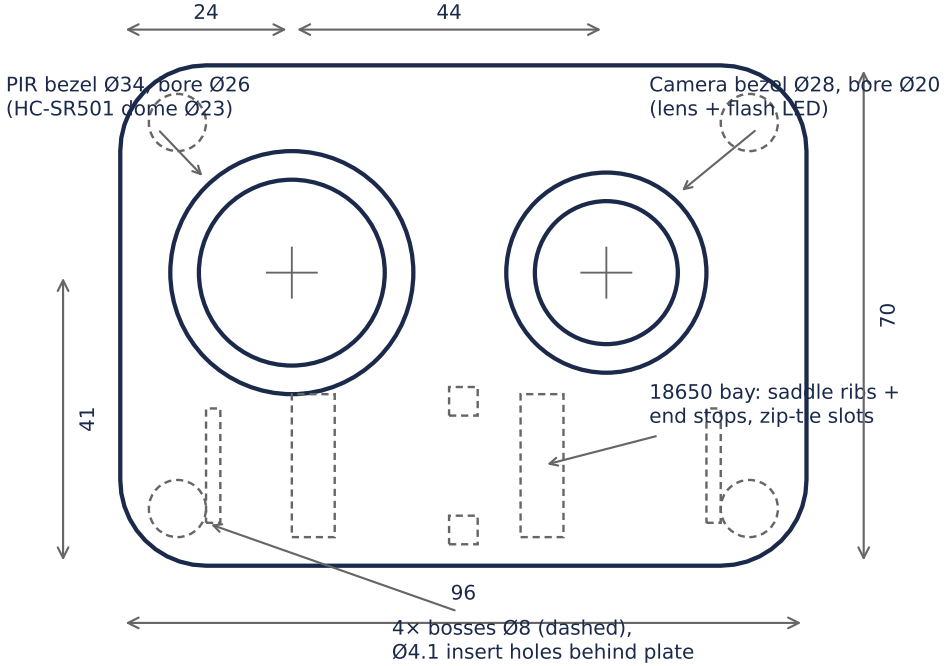
# WhatsApp Doorbell Cam — Pebble Enclosure

Rev 2026-10-08

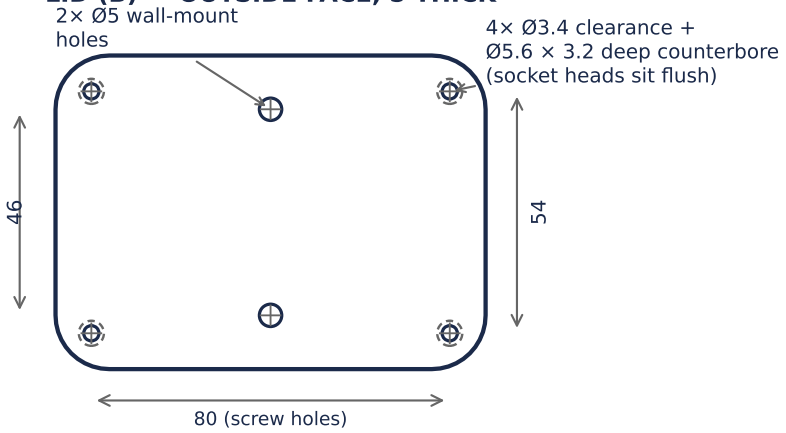
Dimensioned views · Body (A) and Lid (B) · STLs in the case folder

Units: mm · JakinsCraft

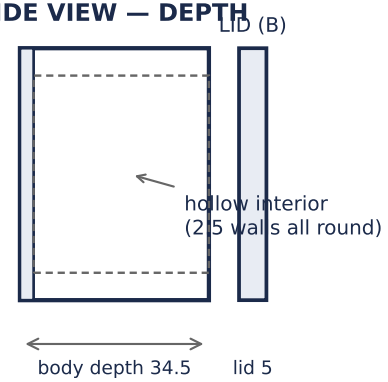
## FRONT VIEW — BODY (A)




## LID (B) — OUTSIDE FACE, 5 THICK




## SIDE VIEW — DEPTH




## BOTTOM VIEW — USB NOTCH



notches 5.5 tall behind front plate: main USB x 41–55 (14 wide), charger USB x 66–78 (12 wide)

PRINTING
- PETG recommended (porch sun/heat). 0.2 mm layers, 3 walls, 15% infill.
- Body prints FRONT-FACE-DOWN; lid prints flat. No supports needed.
- Both STLs are single watertight solids — they open directly in Bambu Studio (fixed 2026-10-02; the old files were not).

BEFORE A LONG PRINT
- Test-fit your parts: HC-SR501 dome is Ø23 (bore Ø26); ESP32-CAM boards vary slightly between sellers. Print, check, then commit.
- USB power cable routes out through the bottom notch.
- Press the 4 M3 brass inserts into the bosses (≈200°C iron) before wiring anything.

# Assembly — WhatsApp Doorbell Cam

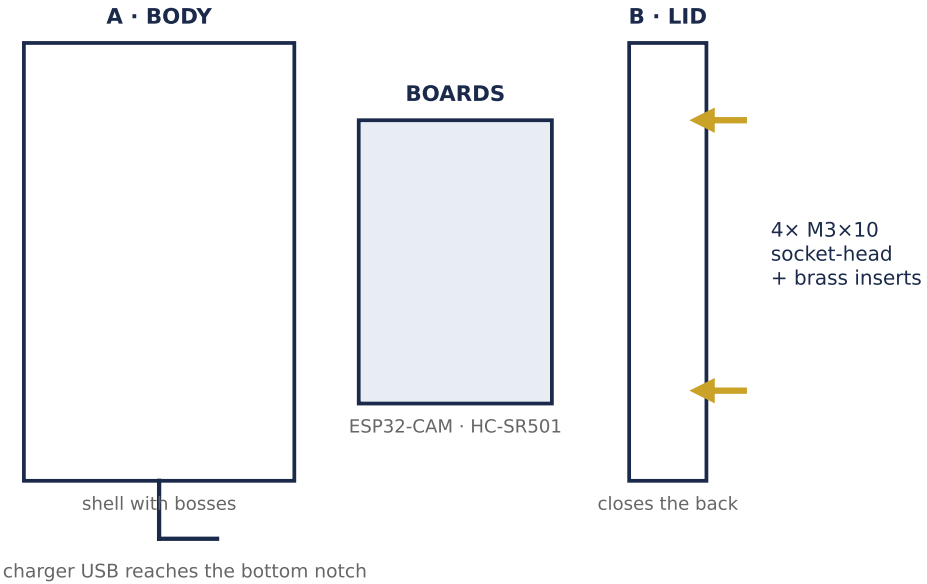
Rev 2026-10-08

Exploded stack, parts list, wiring and build order

Units: mm · JakinsCraft

## EXPLODED VIEW (schematic, front → back)

lens + flash face the Ø20 bore · PIR dome faces the Ø26 bore




## PARTS LIST (* FTDI is for flashing only)

| # | Part | Qty |
|---|---|---|
| A | Enclosure body (3D print, PETG) | 1 |
| B | Enclosure lid (3D print, PETG) | 1 |
| 1 | ESP32-CAM (AI-Thinker) + OV2640 | 1 |
| 2 | HC-SR501 PIR motion sensor | 1 |
| 3 | M3 × 10 socket-head screws | 4 |
| 4 | M3 brass heat-set inserts | 4 |
| 5 | 18650 cell + TP4056 + 5V boost + zip tie | 1 ea. |
| 6 | 470 µF capacitor (across 5V/GND) | 1 |
| 7 | FTDI USB-to-TTL programmer * | 1 |

```
ASSEMBLY ORDER
1. Print A (face-down) and B (flat). No supports.
2. Press 4 brass inserts into the body bosses, flush.
3. Flash the ESP32-CAM with the FTDI before it goes in.
4. Wire the power chain + PIR per the table below;
   470 µF across 5V/GND, close to the ESP32-CAM.
5. Seat the PIR dome in the Ø26 bore, ESP32-CAM lens
   in the Ø20 bore (board long-side horizontal).
   VHB tape or hot glue.
6. Lay the 18650 in the cradle; zip tie up through one
   floor slot, over the cell, down the other slot.
7. Tape the TP4056 + boost beside the cradle; line the
   charger USB up with its bottom notch.
8. Bench-test: wave at the PIR → WhatsApp photo.
9. Fix the lid (B) to the wall through the Ø5 holes —
   counterbore recesses face OUT.
10. Close the body onto the lid with 4× M3×10;
    heads sit flush in the counterbores.
```

## WIRING

| From | To | Note |
|---|---|---|
| 18650 +/− | TP4056 B+/B− | check polarity first |
| TP4056 OUT | boost IN | — |
| boost +5V | ESP32-CAM 5V | 470 µF across 5V/GND |
| GND | common everywhere | battery −, PIR, ESP |
| PIR VCC | 5V rail | — |
| PIR OUT | ESP32-CAM GPIO 13 | wake-up trigger |

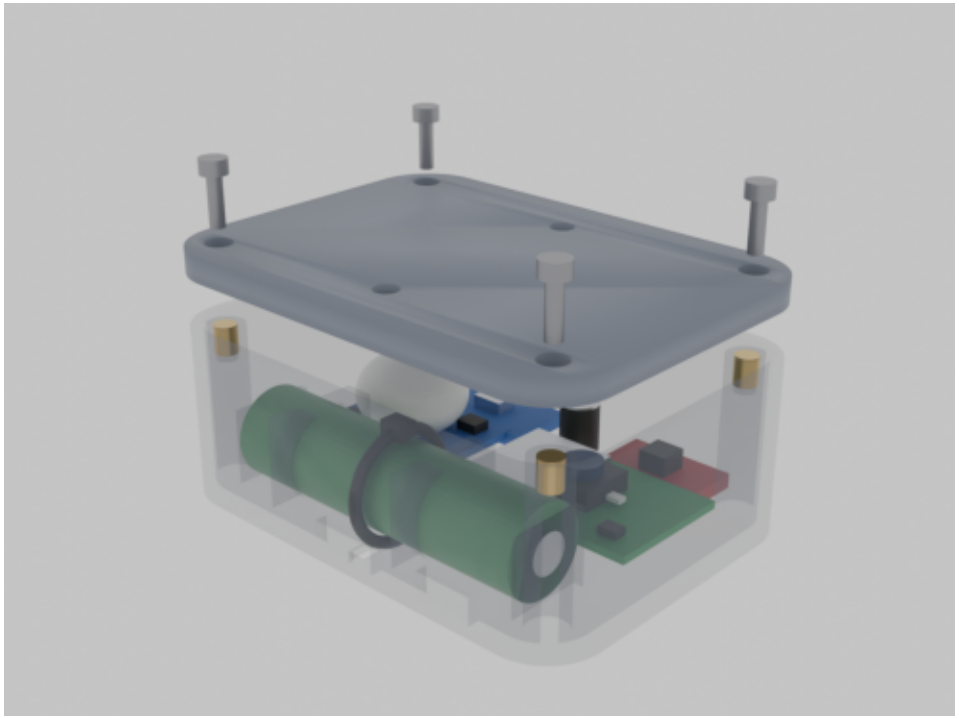
Fit check: AI-Thinker ESP32-CAM ≈ 40.5 × 27 mm · HC-SR501 board ≈ 32 × 24 mm (typical — verify yours)

# Pictorial assembly — WhatsApp Doorbell Cam

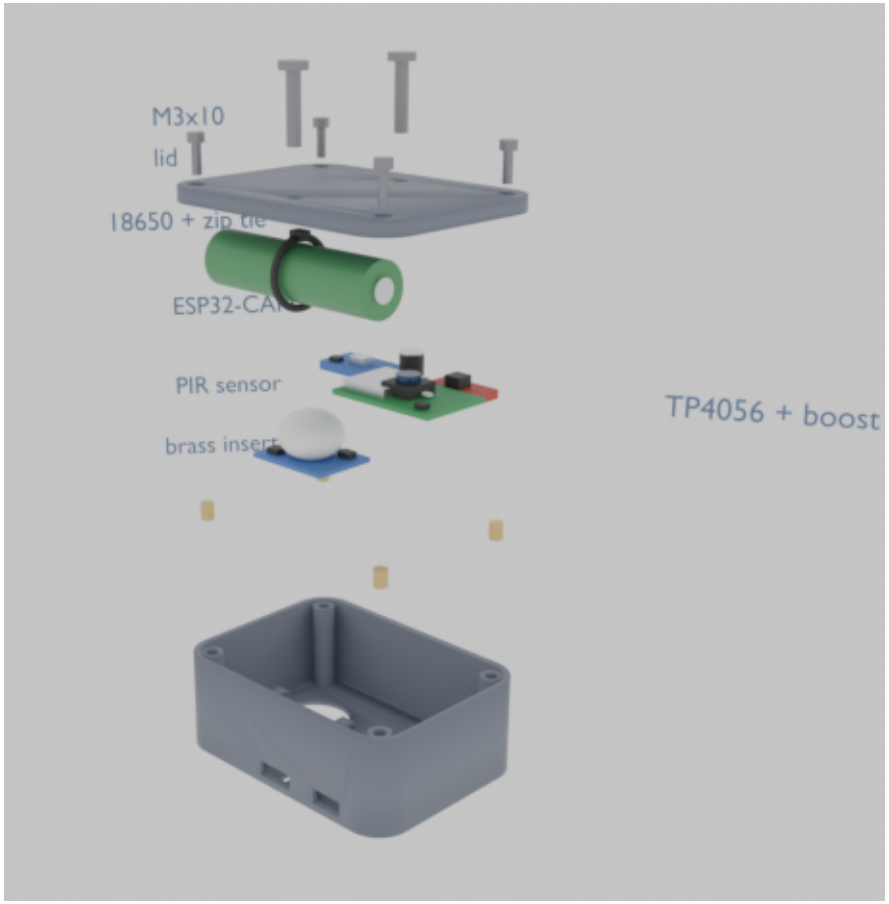
Rev 2026-10-08

Realistic renders · Blender Cycles · true case STLs + modelled components

Units: mm · JakinsCraft



FITTED — transparent body, every component in place,
lid raised with the M3×10 screws above it




EXPLODED — assembly order, bottom to top:
body → brass inserts → PIR → ESP32-CAM → battery → lid → screws

Component models: ESP32-CAM (40.5×27), HC-SR501 (32×24), 18650 cell,
TP4056 charger, 5V boost, 470µF capacitor, zip tie, brass inserts.